MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3130 FOUNDATION DESIGN OF TRANSFER POINT TP-3

DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1005

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1 INTENT 4

2 EXECUTIVE SUMMARY 4

3 SCOPE 4

4 CONCRETE STRUCTURE 4

4.1 Design 4
4.2 Materials 4
4.3 Reference 4

5 INPUT LOADS FROM SUPERSTRUCTURE 4-8

5.1 Support Reaction at Grid 3130/H - 3130/1 4
5.2 Support Reaction at Grid 3130/H - 3130/2 5
5.3 Support Reaction at Grid 3130/E - 3130/1 5
5.4 Support Reaction at Grid 3130/E - 3130/2 5
5.5 Load Combinations 6

6 GEOMETRY

6.1 Geometry of TP-3 Superstructure 7
6.2 Geometry of TP-3 Foundation 7

7 DESIGN OF CONCRETE COMPONENTS 8-14

7.1 Design summary of Pedestals 8
7.2 Design of footings 8
7.3 Design of Tie Beams 8

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING 15

ATTACHMENT 2 STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE DETAIL 18

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE OF TP-3 FOUNDATION 20

ATTACHMENT 4 COMMENT RESOLUTION SHEET 45

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of TP-3
TP3 is the transfer point for conveyor 3130-FDR-004 and conveyor 3130-CNV-001

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of TP-3. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column reinforcement design has been performed in Staad-Pro Software. The footing and tie beam design have been performed in MS Excel.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

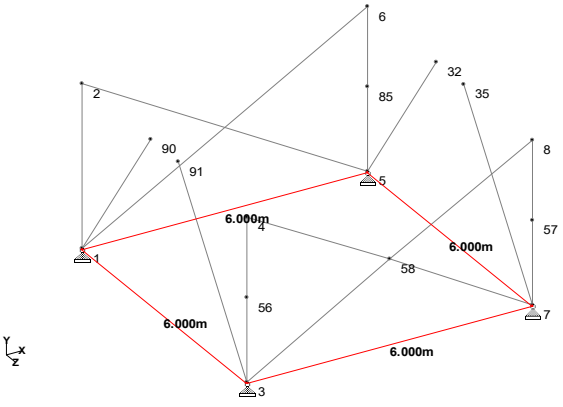
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3130-EG-ST-CAL-1006

5.1 Support Reaction at Grid 3130/H - 3130/1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	1.7	13.7	-2.0
	2 LIVE LOAD	4.1	38.4	5.3
	3 SDL1 - EQUIPMENT DEAD	6.3	22.4	-22.6
	4 SDL2 - EQUIPMENT DEAD	5.4	-1.3	-24.2
	5 WL IN +X	-10.1	-13.9	-1.6
	6 WL IN -X	10.1	13.9	1.6
	7 WL IN +Z	0.0	-6.4	-6.3
	8 WL IN -Z	0.0	6.4	6.3
	9 TEMERATURE LOAD 1	61.3	4.8	2.5
	10 TEMERATURE LOAD 2	-61.3	-4.8	-2.5
	11 EARTHQUAKE LOAD (+X)	-2.3	-3.6	-0.6
	12 EARTHQUAKE LOAD (+Z)	0.3	-2.4	-2.0



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5.2 Support Reaction at Grid 3130/H - 3130/2

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
5	1 DEAD LOAD	-1.7	23.5	-3.0
	2 LIVE LOAD	-4.0	52.0	-0.3
	3 SDL1 - EQUIPMENT DEAD	-17.6	38.9	-31.7
	4 SDL2 - EQUIPMENT DEAD	-13.9	-2.3	-36.1
	5 WL IN +X	-8.9	13.9	-2.1
	6 WL IN -X	8.9	-13.9	2.1
	7 WL IN +Z	-0.6	-13.4	-10.4
	8 WL IN -Z	0.6	13.4	10.4
	9 TEMERATURE LOAD 1	-57.4	-4.8	-0.9
	10 TEMERATURE LOAD 2	57.4	4.8	0.9
	11 EARTHQUAKE LOAD (+X)	-2.1	3.6	-0.2
	12 EARTHQUAKE LOAD (+Z)	-0.3	-3.9	-2.8

5.3 Support Reaction at Grid 3130/E - 3130/1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
3	1 DEAD LOAD	0.1	57.0	1.4
	2 LIVE LOAD	2.4	57.1	-5.2
	3 SDL1 - EQUIPMENT DEAD	1.4	129.1	-33.6
	4 SDL2 - EQUIPMENT DEAD	0.3	119.5	-29.6
	5 WL IN +X	-14.8	-16.4	-3.5
	6 WL IN -X	14.8	16.4	3.5
	7 WL IN +Z	0.4	6.4	-6.7
	8 WL IN -Z	-0.4	-6.4	6.7
	9 TEMERATURE LOAD 1	39.1	-4.8	1.3
	10 TEMERATURE LOAD 2	-39.1	4.8	-1.3
	11 EARTHQUAKE LOAD (+X)	-2.5	-2.5	-0.8
	12 EARTHQUAKE LOAD (+Z)	0.0	2.7	-2.7

5.4 Support Reaction at Grid 3130/E - 3130/2

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
7	1 DEAD LOAD	-3.9	71.8	3.7
	2 LIVE LOAD	-2.5	59.6	0.2
	3 SDL1 - EQUIPMENT DEAD	-12.1	180.7	-41.1
	4 SDL2 - EQUIPMENT DEAD	-10.8	166.1	-36.0
	5 WL IN +X	-14.6	16.4	7.1
	6 WL IN -X	14.6	-16.4	-7.1
	7 WL IN +Z	0.3	13.4	-11.0
	8 WL IN -Z	-0.3	-13.4	11.0
	9 TEMERATURE LOAD 1	-43.0	4.8	-3.0
	10 TEMERATURE LOAD 2	43.0	-4.8	3.0
	11 EARTHQUAKE LOAD (+X)	-2.4	2.5	1.6
	12 EARTHQUAKE LOAD (+Z)	0.1	4.2	-3.7

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5.5 LOAD COMBINATIONS

5.5.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD 301 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2)	
LOAD 303 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 304 (DL + SDL2 + TL1) + LL	
LOAD 305 (DL + SDL1 + TL2) + LL	
LOAD 306 (DL + SDL2 + TL2) + LL	
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X)	
LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z)	
LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z)	
LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X)	
LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X)	
LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z)	
LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X)	
LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z)	
LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z)	
LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X)	
LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X)	
LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z)	
LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	

5.5.2 Load Combination for Concrete design

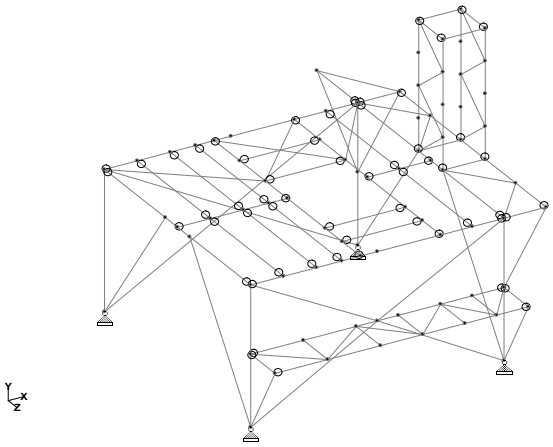
Load Case	Reference
LOAD 401 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL + SDL2)	
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X)	
LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z)	
LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X)	
LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X)	
LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z)	
LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X)	
LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z)	
LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z)	
LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X)	
LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X)	
LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z)	
LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z)	
LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X)	
LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z)	
LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z)	
LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X)	
LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X)	
LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z)	
LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	

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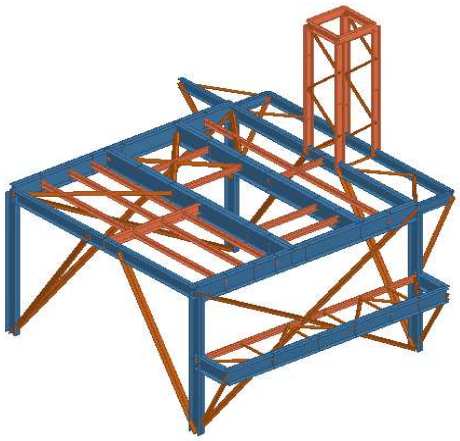
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6.2 Geometry of TP-3 Superstructure

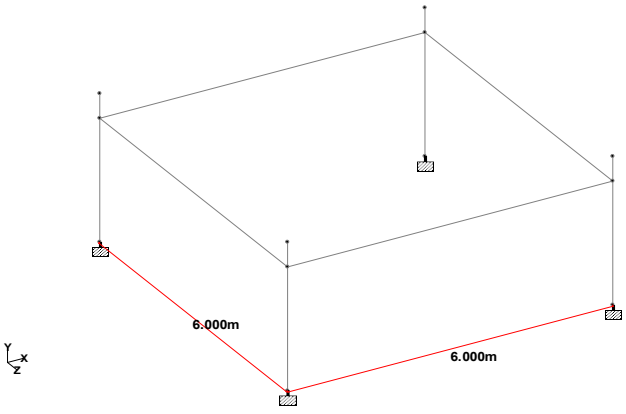


Staad Model of TP-3 Superstructure

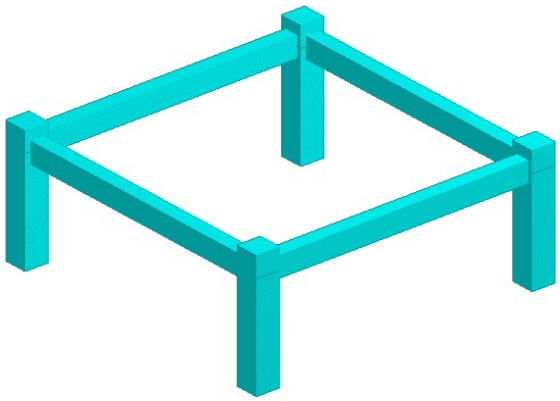


Isometric view of TP-3 Superstructure

6.3 Geometry of TP-3 Foundation



Staad Model of TP-3 Foundation framework



Isometric view of TP-3 Foundation framework

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
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7. DESIGN OF CONCRETE COMPONENTS

All footings & tie beams have been designed using worksheet in accordance to SBC-304.

All columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachment 3 - Staad Output file.

7.1 Design summary of pedestals

7.1.1 Pedestal Reinforcement Summary

Column mark	Dimension (mm)	Reinf. Area Required (mm^2)	Reinf. Area Provided (mm^2)	Reinforcement	Links	Remarks	Numbers
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/H-Grid 3130/1	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/H-Grid 3130/2	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/E-Grid 3130/1	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/E-Grid 3130/2	1

7.2 Design of Footings

7.2.1 Footing marking, Maximum Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (mm)	Width (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
Column at 3130/H - 3130/1	F1	2300	2300	400	92	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.18	1
Column at 3130/H - 3130/2	F1	2300	2300	400	110	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.21	1
Column at 3130/E - 3130/1	F1	2300	2300	400	140	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.27	1
Column at 3130/E - 3130/2	**F1**	**2300**	**2300**	**400**	**172**	**D12 @ 150c/c B/W**	**D12 @ 150c/c B/W**	**0.33**	**1**

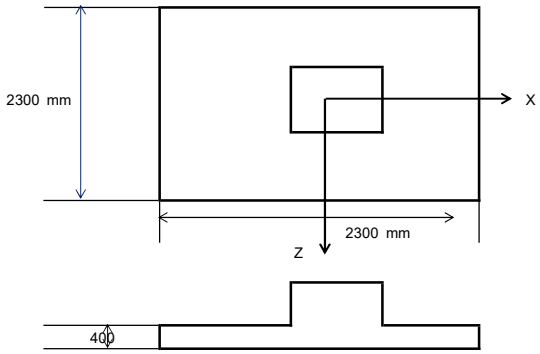
The design of the most critical foundation of the group is attached in the next pages.

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Project Country	**KINGDOM OF SAUDI ARABIA**

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.2.2 Design of footing F1 at Grid 3130/E-3130/2

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	145.15 KN	*inputs
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	50.78 KN	
Top of footing (staad support)	T.O.F	=	-1.600	m	Uplift	=	0.00 KN	
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0 KN	
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	37.0 KN/m^2	
Length of footing	Lz	=	2.30	m				
Width of footing	Lx	=	2.30	m				
Thickness of footing	D	=	0.40	m				
Depth of pedestal	Bx	=	0.50	m				
Width of pedestal	Bz	=	0.50	m				
Height of Load point from bottom of footing	H	=	0.400	m				
Section Modulus of Pedestal	Zxx	=	2.03	m^3				
Section Modulus of Pedestal	Zzz	=	2.03	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	12	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	10.4	306.01	40.8	90.56	-0.3	-21.0	501.9	149.92	39.85	1	0.28
302	10.6	291.19	41.5	90.75	-0.27	-20.9	487.1	147.16	37.01	1	0.27
303	2.3	371.54	42.4	94.63	-0.34	-15.2	567.5	161.46	53.09	1	0.30
304	2.4	356.72	43.1	94.82	-0.31	-15.1	552.7	158.70	50.24	1	0.30
305	17.5	359.73	39.3	86.80	-0.25	-25.9	555.7	160.62	49.47	1	0.30
306	17.7	344.92	40.0	86.98	-0.2	-25.8	540.9	157.86	46.62	1	0.29
307	21.0	366.2	39.8	87.16	0.09	-46.5	562.1	172.19	40.34	1	0.32
308	-0.9	335.26	42.0	94.19	-0.68	5.1	531.2	149.40	51.43	1	0.28
309	9.8	363.37	48.8	109.38	-0.48	-20.1	559.3	169.57	41.89	1	0.32
310	10.3	338.09	33.0	71.97	-0.11	-21.3	534.0	146.94	54.96	1	0.27
311	21.2	351.38	40.4	87.35	0.1	-46.4	547.3	169.43	37.49	1	0.32
312	-0.8	320.44	42.6	94.38	-0.65	5.2	516.4	146.74	48.48	1	0.27
313	9.9	348.55	49.4	109.57	-0.45	-20.0	544.5	166.82	39.04	1	0.31
314	10.4	323.27	33.6	72.16	-0.1	-21.2	519.2	144.18	52.11	1	0.27
315	11.4	352.38	40.6	89.97	-0.25	-23.8	548.3	159.74	47.56	1	0.30
316	8.7	349.08	41.1	91.38	-0.34	-17.6	545.0	156.78	49.28	1	0.29
317	10.0	353.51	42.5	94.63	-0.32	-20.7	549.4	160.72	47.01	1	0.30
318	10.1	347.95	39.2	86.72	-0.27	-20.7	543.9	155.80	49.83	1	0.29
319	11.5	337.56	41.3	90.16	-0.2	-23.7	533.5	156.98	44.72	1	0.29
320	8.9	334.27	41.8	91.57	-0.31	-17.5	530.2	154.02	46.43	1	0.29
321	10.2	338.7	43.2	94.82	-0.29	-20.6	534.6	157.97	44.16	1	0.29
322	10.2	333.13	39.9	86.91	-0.24	-20.6	529.1	153.04	46.99	1	0.29
323	20.9	204.24	23.0	49.65	0.34	-47.1	400.2	123.34	27.95	1	0.23
324	-8.4	162.98	26.0	59.03	-0.7	21.8	358.9	107.73	27.97	1	0.20
325	5.9	200.46	35.0	79.28	-0.42	-11.8	396.4	119.85	30.01	1	0.22
326	6.6	166.76	13.9	29.41	0.06	-13.4	362.7	89.69	47.44	1	0.17
327	21.0	195.34	23.4	49.76	0.35	-47.0	391.3	121.69	26.24	1	0.23
328	-8.3	154.09	26.4	59.14	-0.7	21.9	350.0	106.13	26.21	1	0.20
329	6.0	191.57	35.4	79.40	-0.4	-11.7	387.5	118.20	28.31	1	0.22
330	6.7	157.86	14.3	29.52	0.08	-13.4	353.8	88.03	45.73	1	0.16
331	8.0	185.8	24.2	53.39	-0.12	-16.7	381.7	106.74	37.58	1	0.20
332	4.5	181.41	24.8	55.29	-0.24	-8.5	377.3	102.80	39.87	1	0.19
333	6.2	187.32	26.7	59.63	-0.2	-12.6	383.3	108.05	36.84	1	0.20
334	6.3	179.9	22.3	49.06	-0.15	-12.7	375.8	101.48	40.61	1	0.19
335	8.1	176.91	24.6	53.51	-0.1	-16.7	372.8	105.09	35.87	1	0.20
336	4.6	172.52	25.2	55.40	-0.22	-8.5	368.5	101.14	38.16	1	0.19
337	6.3	178.43	27.1	59.74	-0.2	-12.5	374.4	106.40	35.14	1	0.20
338	6.4	171.01	22.7	49.17	-0.1	-12.6	366.9	99.83	38.90	1	0.19

Max P_{max}	=	172.19	KN/m^2
Max P_{min}	=	26.21	KN/m^2
Resultant F.O.S	=	0.32	

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	14.6	428.4	57.1	126.80	-0.4	-29.45	158.04	3.94
402	14.8	407.7	58.1	127.06	-0.38	-29.30	154.17	-0.04
403	3.1	538.1	59.4	132.51	-0.47	-21.21	177.52	25.90
404	3.2	517.3	60.4	132.78	-0.43	-21.08	173.66	21.92
405	24.4	521.5	55.1	121.55	-0.35	-36.13	176.34	20.83
406	24.6	500.8	56.0	121.81	-0.31	-36.00	172.49	16.84
407	35.7	430.0	46.7	101.25	0.47	-80.13	170.74	-8.15
408	-11.2	364.0	51.4	116.26	-1.19	30.11	140.99	-3.36
409	11.7	424.0	65.9	148.65	-0.74	-23.72	165.15	-4.85
410	12.8	370.1	32.1	68.86	0.03	-26.31	116.88	23.03
411	35.8	412.3	47.5	101.47	0.51	-80.02	167.43	-11.57
412	-11.0	346.2	52.2	116.48	-1.15	30.23	137.80	-6.90
413	11.8	406.2	66.7	148.88	-0.71	-23.60	161.84	-8.27
414	13.0	352.3	32.9	69.08	0.06	-26.19	113.57	19.61
415	14.8	400.2	48.5	107.41	-0.27	-30.88	143.84	7.45
416	9.8	393.9	49.5	110.10	-0.44	-19.14	138.20	10.72
417	12.2	402.3	52.2	116.30	-0.41	-24.94	145.71	6.40
418	12.3	391.7	45.9	101.21	-0.31	-25.08	136.33	11.77
419	14.9	382.4	49.3	107.63	-0.24	-30.76	140.53	4.04
420	9.9	376.1	50.3	110.32	-0.41	-19.03	134.89	7.31
421	12.4	384.5	53.0	116.52	-0.37	-24.82	142.39	2.99
422	12.4	374.0	46.7	101.43	-0.27	-24.97	133.02	8.36
423	32.8	308.4	34.4	74.01	0.56	-74.05	131.32	-14.71
424	-14.0	242.4	39.1	89.02	-1.1	36.20	107.57	-15.92
425	8.8	302.4	53.6	121.41	-0.66	-17.63	125.73	-11.41
426	10.0	248.5	19.9	41.61	0.11	-20.23	77.46	16.47
427	32.9	295.1	35.0	74.18	0.58	-73.96	128.83	-17.27
428	-13.9	229.1	39.7	89.19	-1.07	36.28	105.18	-18.57
429	8.9	289.0	54.2	121.58	-0.63	-17.55	123.25	-13.97
430	10.1	235.1	20.5	41.78	0.14	-20.14	74.98	13.91
431	11.9	278.6	36.3	80.16	-0.19	-24.80	104.42	0.89
432	6.9	272.3	37.2	82.86	-0.36	-13.06	98.77	4.17
433	9.4	280.7	39.9	89.06	-0.32	-18.86	106.28	-0.16
434	9.4	270.1	33.6	73.97	-0.22	-18.99	96.90	5.22
435	12.0	265.2	36.9	80.34	-0.16	-24.71	101.94	-1.67
436	7.0	258.9	37.8	83.03	-0.33	-12.97	96.29	1.60
437	9.5	267.4	40.5	89.23	-0.3	-18.77	103.80	-2.72
438	9.5	256.8	34.2	74.14	-0.2	-18.91	94.43	2.65

Max Net factored B.P = 177.52 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z Muz = 71.89 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 603.5 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 574.2 mm^2

A_{req} = 603.5 mm^2

Required Spacing = **187.4** mm

Length of cantilever along Z = 0.9 m

Moment about X Mux = 71.89 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 627.8 mm^2

Effective Depth Along X = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 552.6 mm^2

A_{req} = 627.8 mm^2

Required Spacing = **180.1** mm

Provide Bottom Reinforcement = **12 mm dia @ 150 c/c**

Ast prov = 754.0 mm^2

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Top Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: M_{uz} = 0.00 KNm

$$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

= 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

= 0.0 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 287.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Length of cantilever along Z = 0.9 m

Moment about X: M_{ux} = 0.00 KNm

$$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

= 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

= 0.0 mm^2

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

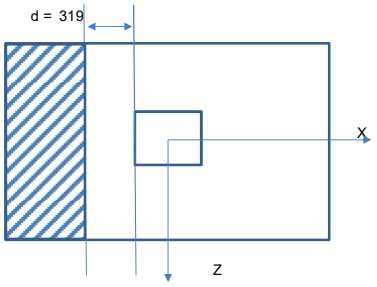
A_{stmin} = 276.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Provide top Reinforcement = **12 mm dia @ 150 c/c** Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever L = 0.581 m

Reinforcement ratio ρ_w = 0.0024

Effective Depth Along X d = 319 mm

Shear Force at d distance V_u = 103.14 KN

Allowable shear strength φVc = 236.08 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} b_w d + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 29.96 KNm

φ = 0.75 for shear

φVc = 237.02 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} b_w d + 17\rho_w\right) b_w d$

φVc = 236.080808 KN *Table 22.5.5.1 SBC 304*

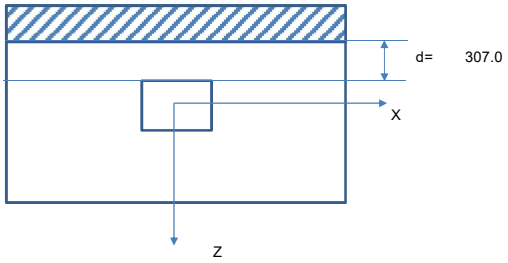
iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 410.472 KN

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023	Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT	Revision Tex	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

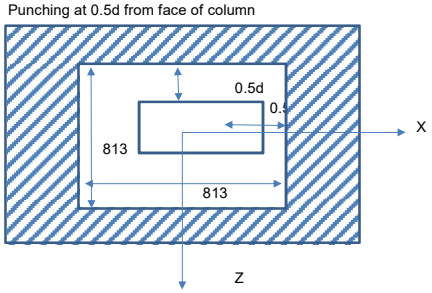
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.593	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0025						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	307	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	105.27	KN				Mu	= 31.21 KNm
Allowable shear strength		φVc	=	227.56	KN	O.K			φ	= 0.75 for shear
									φVc	= 227.90 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 227.561653 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 395.03 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	5.29	m2
Punching area		=	0.66	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3252.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.87 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	Vc	=	$v_c b_0 d$
Depth at section	d	=	313	mm		=	1987.21 KN
Shear Force at section	Vu	=	821.7	KN	φ	=	0.75 for shear
Allowable shear strength	φVc	=	1490.40	KN O.K	φV_n	=	φVc
					φVc	=	1490.4 KN

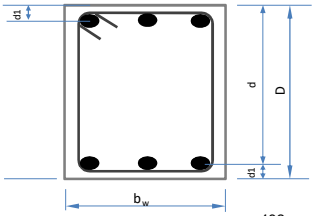
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3 Design of Tie Beams

7.3.1 Beam marked TB1

Staad beam number - 123,125,127,128



Width of beam	=	**300**	mm		
Depth of beam	=	**500**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	432	mm		
Dia of longitudinal bar	=	**16**	mm	d =	432 mm
Dia of stirrup bar	=	**10**	mm	d' =	68 mm
				h =	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	0	Staad beam number -	0	=	**55**	KNm
B.M at span	Mz	L/C -	0	Staad beam number -	0	=	**31**	KNm
Shear Force	Fy	L/C -	0	Staad beam number -	0	=	**30**	KN

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

Singly reinforced beam

$\rho_{max} = \frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 55 KNm

Mu_2 = Mu - Mu_{lim} = 55 - 382.3 = 0.00 KNm

$A_{st1} = \frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

$A_{st2} = \frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 55 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 342.8 mm²

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 0 mm²

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

e_{sc} = 0.0017

Asc = 0.0000 mm²

= 456.4 mm² or 432 mm²

= 456.4 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0.00 + 0

= 456.4 mm²

Top layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm² Safe

ρ_w = 0.00465421

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 31 KNm

Mu_2 = Mu - Mu_{lim} = 31 - 382.3

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 31 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 191.6 mm^2

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 0 mm^2

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 456.4 mm^2 or 432 mm^2 = 456.383298 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0 + 0

= 456.4 mm^2

bottom layer1	3	nos	16	mm	dia	+		nos		mm	dia
bottom layer2		nos		mm	dia	+		nos		mm	dia

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.004654211

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 125.1 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 132.9 KN

V_c = $0.29\lambda\sqrt{f_c'} b_w d$ = 222.3 KN

fV_c = 0.75 x 125.1 = 93.82 KN > 30 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((30-93.9) x 10^3)/(0.75 x 420 x 432) = 0 mm^2/mm

No. of legs = 2 Spacing provided = 200 mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s	=	0 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s	=	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s	=	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	216 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

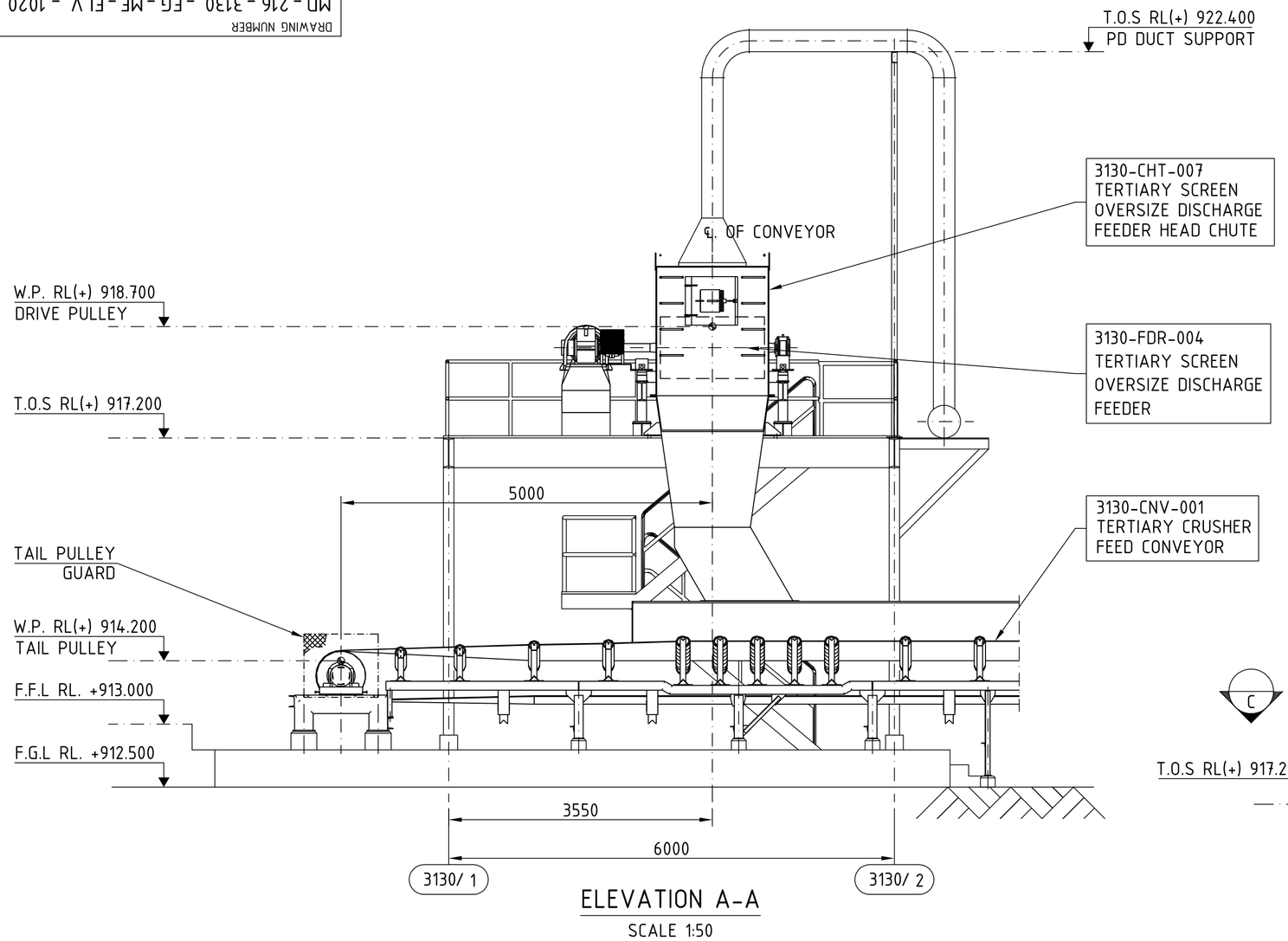
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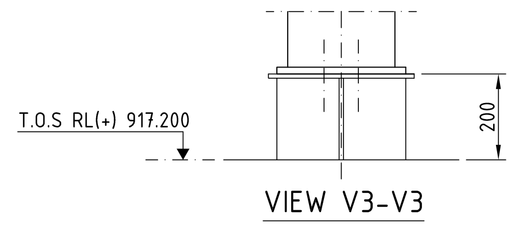
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ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

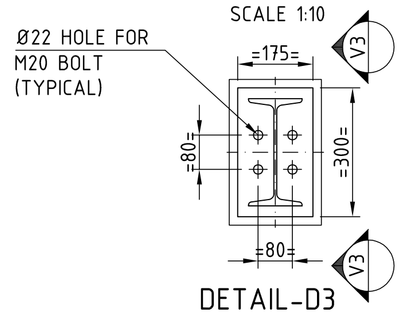
DRAWING NUMBER MD-216-3130-EG-ME-ELV-1020 REV D01



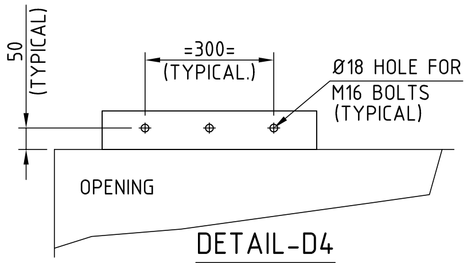
ELEVATION A-A
SCALE 1:50



VIEW V3-V3
SCALE 1:10



DETAIL-D3
SCALE 1:10

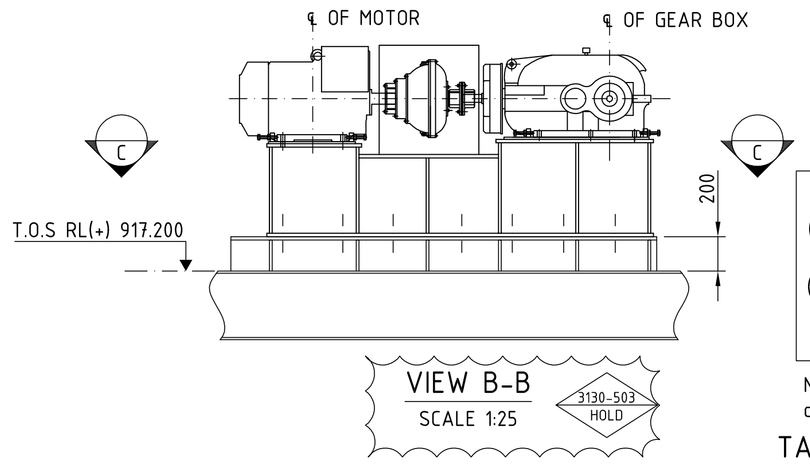


DETAIL-D4
(TYPICAL)
SCALE 1:10

ABBREVIATION

CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



VIEW B-B
SCALE 1:25
3130-503 HOLD

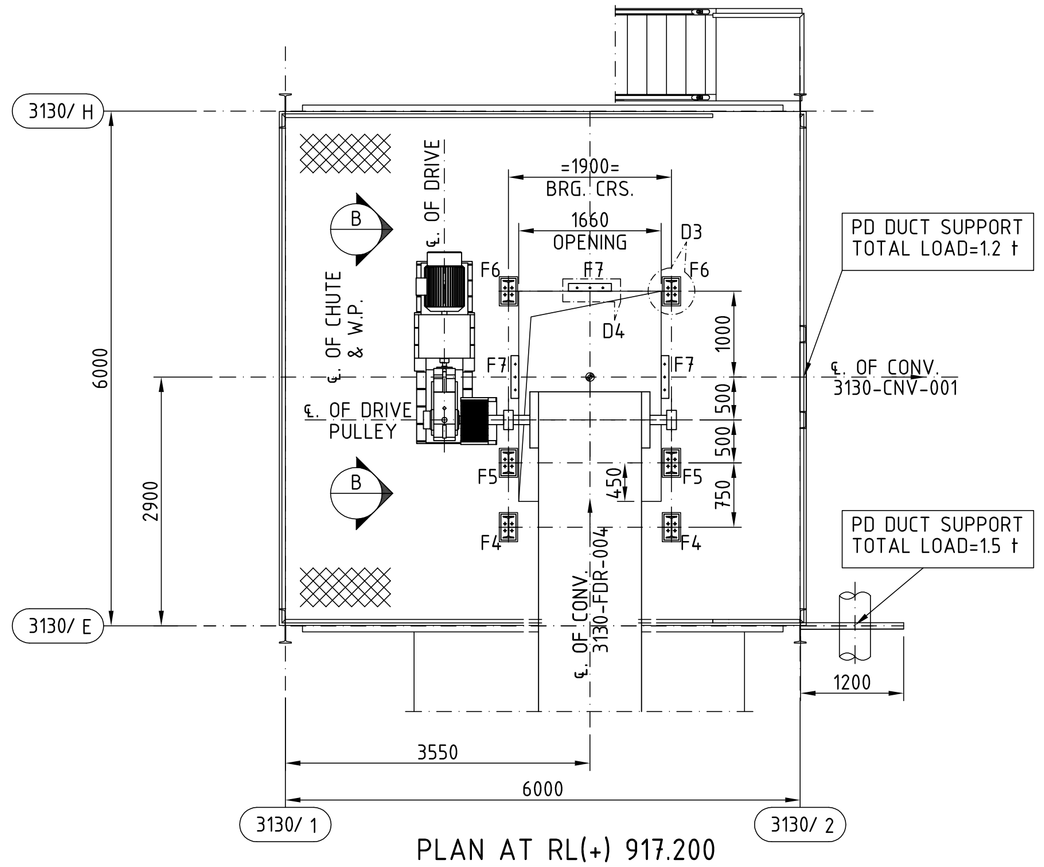
SIGN CONVENTION

(V) VERTICAL LOAD	: UPWARD (-) : DOWNWARD (+)
(H) HORIZONTAL LOAD	: TOWARDS HEAD PULLEY=(+) : TOWARDS TAIL PULLEY=(-) : REVERSIBLE LOAD=(±)

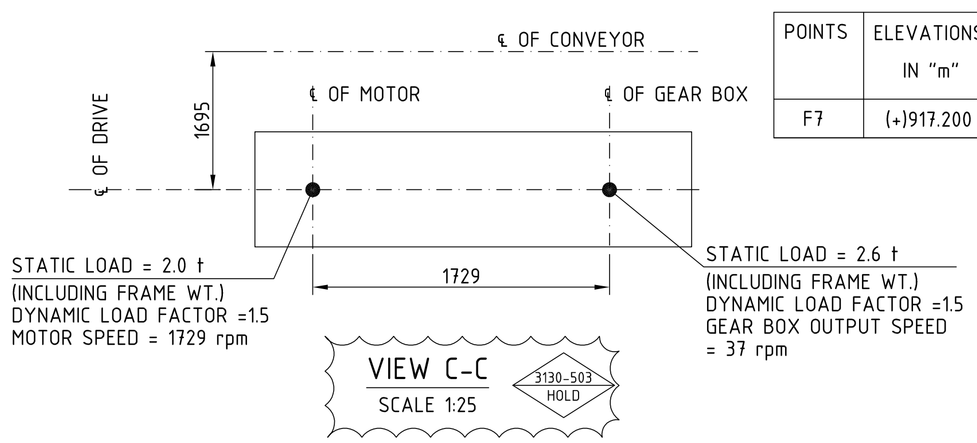
Note: Wind load & seismic to be considered during civil & structural design

TABLE-4 : FOR CHUTE SADDLE LOAD

POINTS	ELEVATIONS IN "m"	V (IN "t")
F7	(+)917.200	(+)1.5



PLAN AT RL(+) 917.200
SCALE 1:50



VIEW C-C
SCALE 1:25
3130-503 HOLD

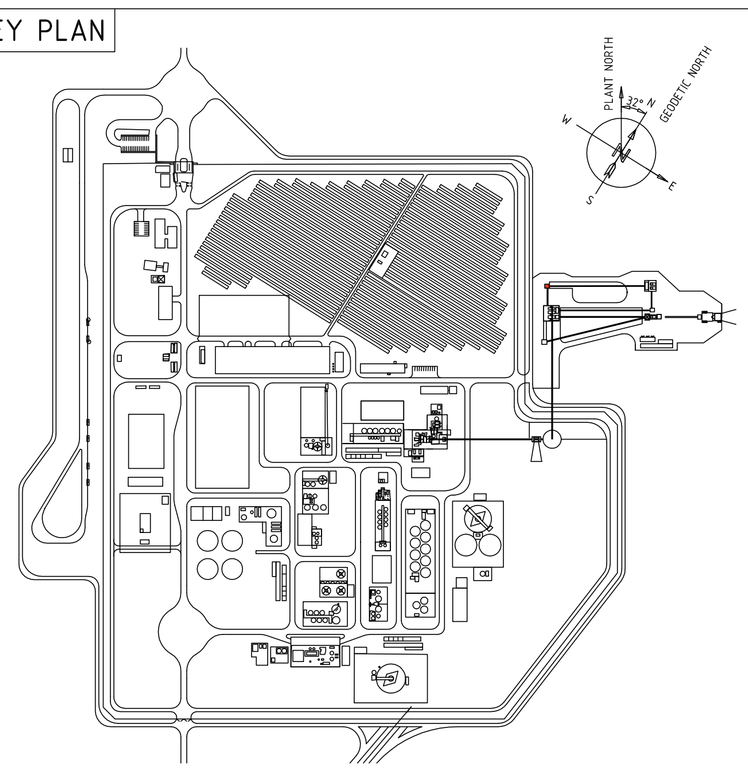
TABLE-3 : FOR DRIVE PULLEY FRAME LOAD

FOR CONV. 3130-FDR-004

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F4	(+)917.200	(+) 4.55	(-) 2.30	(±) 0.45	(+) 5.55	(-) 2.55	(±) 0.55
F5		(+) 3.35	(-) 2.75	(±) 0.35	(+) 1.95	(-) 2.35	(±) 0.20
F6		(+) 2.95	(-) 0.65	(±) 0.30	(-) 1.10	(-) 2.15	(±) 0.20

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	01/11/2023	AS BUILT	SM	BRD	BD	BGR	
		C01	18/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER COMMENTS DT. 21.02.2020 & REVISED AS MARKED	SM	BRD	BD	BGR	
MD-216-3130-EG-ME-LPX-1015	TRANSFER POINT TP#3 - PLAN VIEW	A01	28/01/2020	ISSUED FOR APPROVAL	SM	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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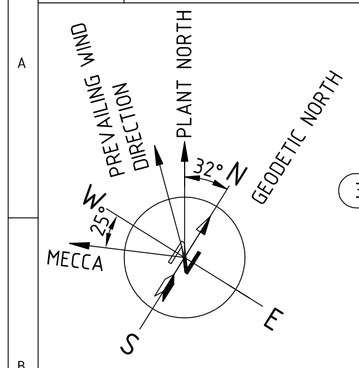
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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZC01 _MTL01 _31379 — DOCUMENT ID: OU100671068 — REVISION: 2

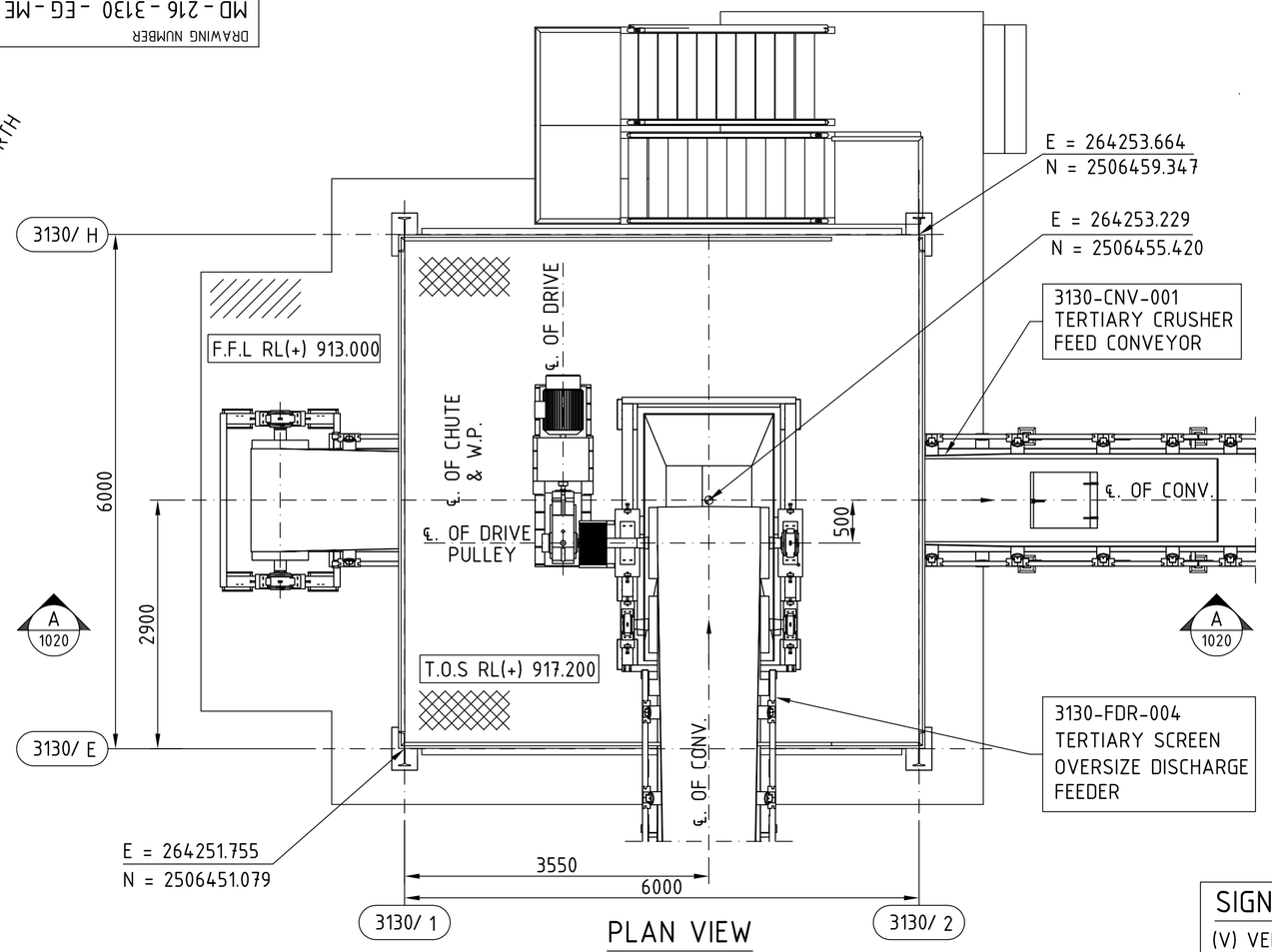
DRAWN	SM	21/01/2020
CHECKED	BRD	22/01/2020
ENGINEER	BD	24/01/2020
APPROVED	BGR	28/01/2020
EMPLOYER		
SCALE	1:50 1:25 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TRANSFER POINT TP#3 - ELEVATION

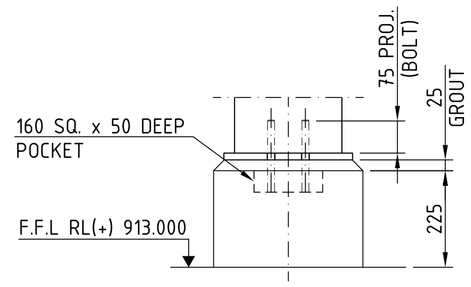
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DRAWING NUMBER MD-216-3130-EG-ME-LPX-1015 REV 01

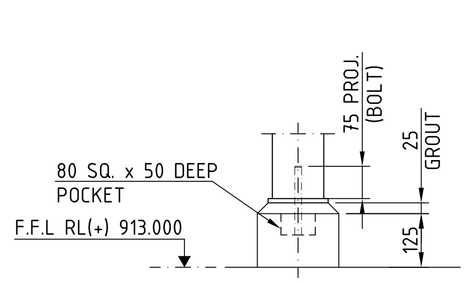




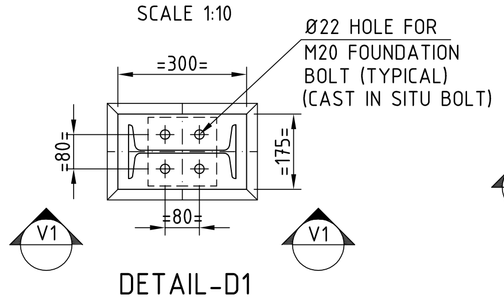
PLAN VIEW
SCALE 1:50



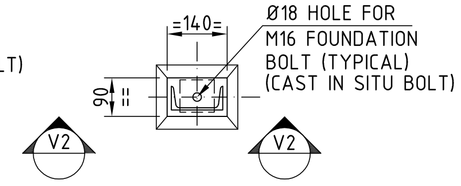
VIEW V1-V1
SCALE 1:10



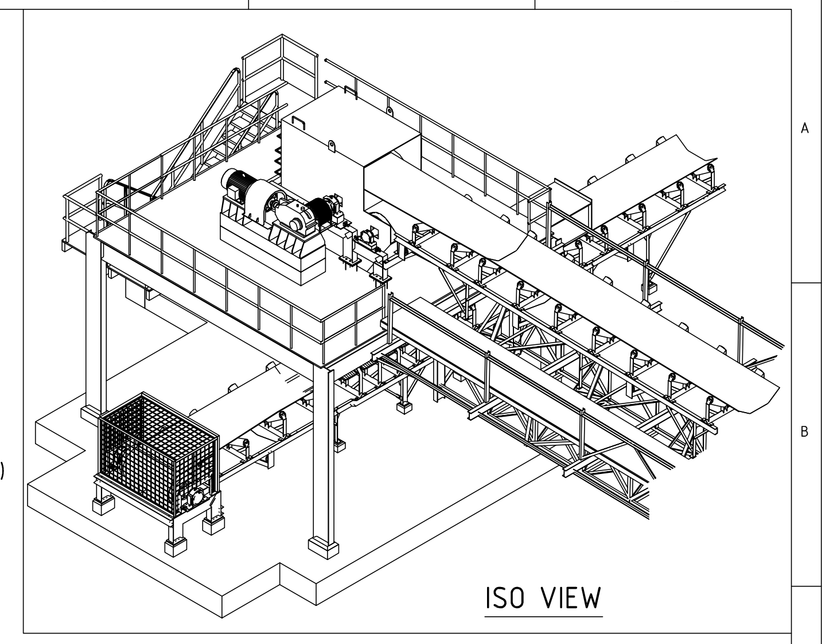
VIEW V2-V2
SCALE 1:10



DETAIL-D1
SCALE 1:10



DETAIL-D2
SCALE 1:10



ISO VIEW

SIGN CONVENTION

(V) VERTICAL LOAD	: UPWARD (-)
	: DOWNWARD (+)
(H) HORIZONTAL LOAD	: TOWARDS HEAD PULLEY=(+)
	: TOWARDS TAIL PULLEY=(-)
	: REVERSIBLE LOAD=(±)

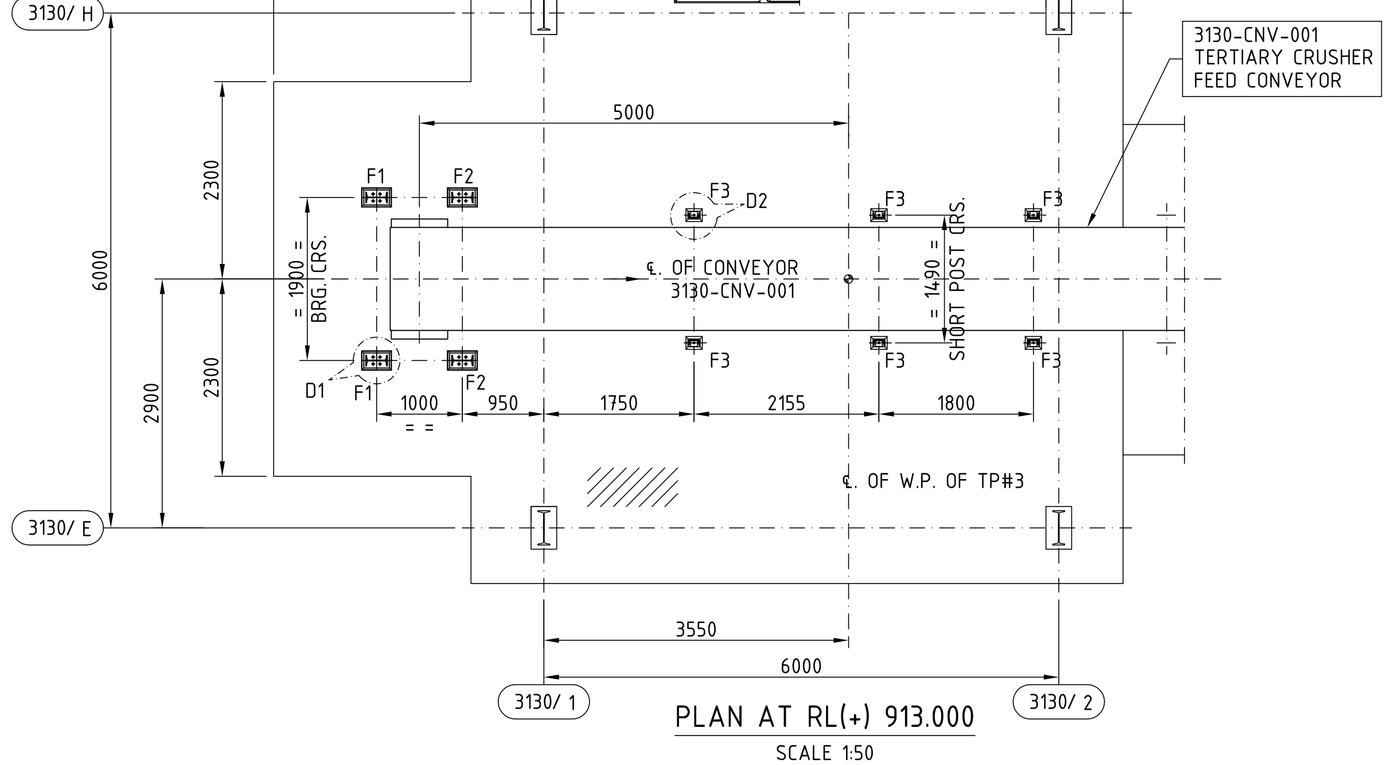
Note: Wind load & seismic to be considered during civil & structural design

ABBREVIATION

CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL



PLAN AT RL(+) 913.000
SCALE 1:50

TABLE-1 : FOR TAIL PULLEY FRAME LOAD

FOR CONV. 3130-CNV-001

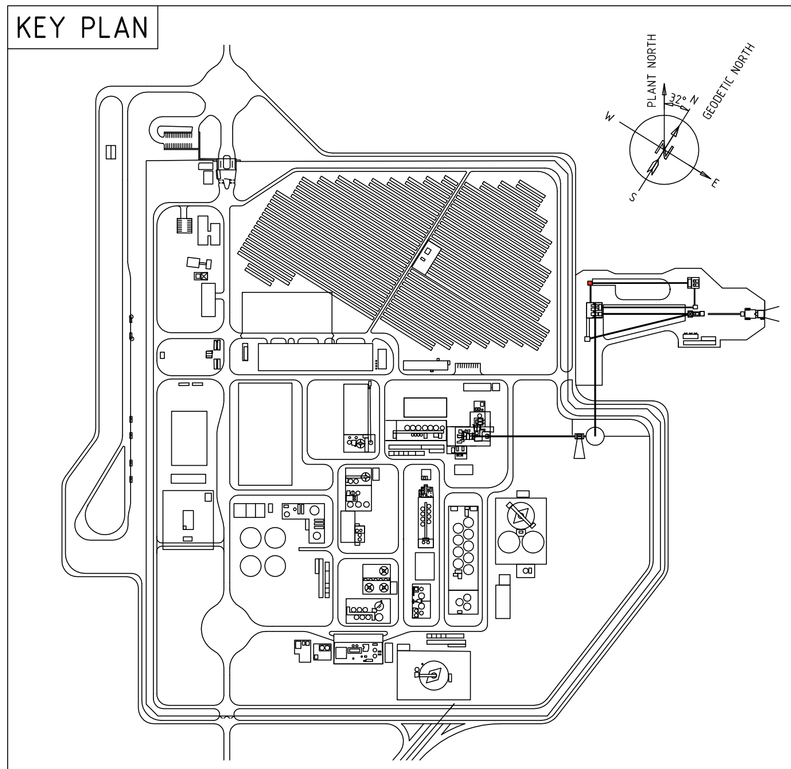
POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN t)	
			ALONG THE CONV. (IN "t")	ACROSS THE CONV. (IN "t")
F1	(+)913.000	(-)2.75	(+)2.10	(±)0.2
F2		(+)5.00	(+)3.00	(±)0.2

TABLE-2 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3130-CNV-001

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F3	(+)913.000	(+) 0.65	(+) 0.85	(±) 0.15	(±) 0.10

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	01-11-2023	AS BUILT	SM	BRD	BD	BGR	
		C01	19/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
				COMMENTS DT. 21.02.2020 & REVISED AS MARKED					
MD-216-3130-EG-ME-ELV-1020	TRANSFER POINT TP#3 - ELEVATION	A01	28/01/2020	ISSUED FOR APPROVAL	SM	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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DRAWN	SM	21/01/2020
CHECKED	BRD	22/01/2020
ENGINEER	BD	24/01/2020
APPROVED	BGR	28/01/2020
EMPLOYER		
SCALE	1:50 1:10	A1

LARSEN & TOUBRO DRAWING NUMBER: O19045_100_H_G_16132_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31378 — DOCUMENT ID: OU100671067 — REVISION: 2

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TRANSFER POINT TP#3 - PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1015 — REV: D01

SM BRD BD BGR

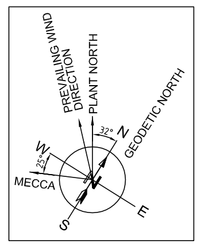
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

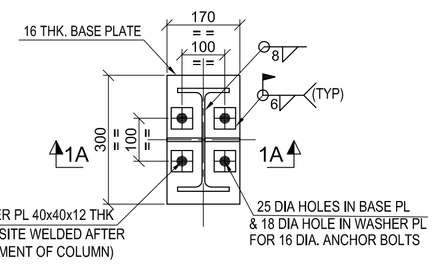
Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

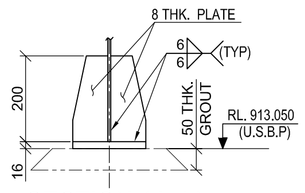
ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE LAYOUT**

DRAWING NUMBER MD-216-3130-EG-ST-GAD-1024 REV A01

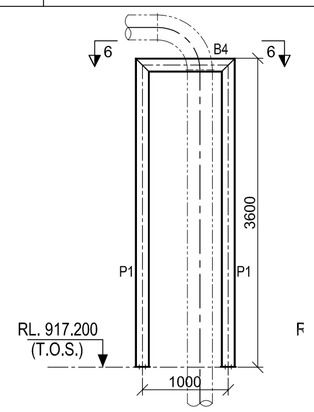




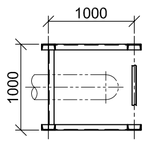
DETAIL OF BASE PLATE MKD. 'BP2'
(SCALE - 1:10)



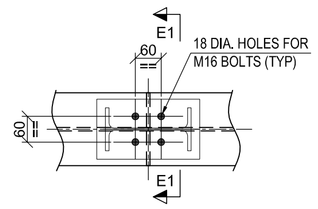
SECTION 1A-1A
(SCALE - 1:10)



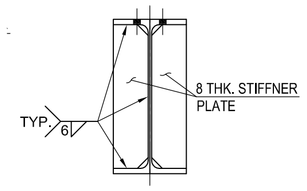
SECTION 5-5
(SCALE-1:50)



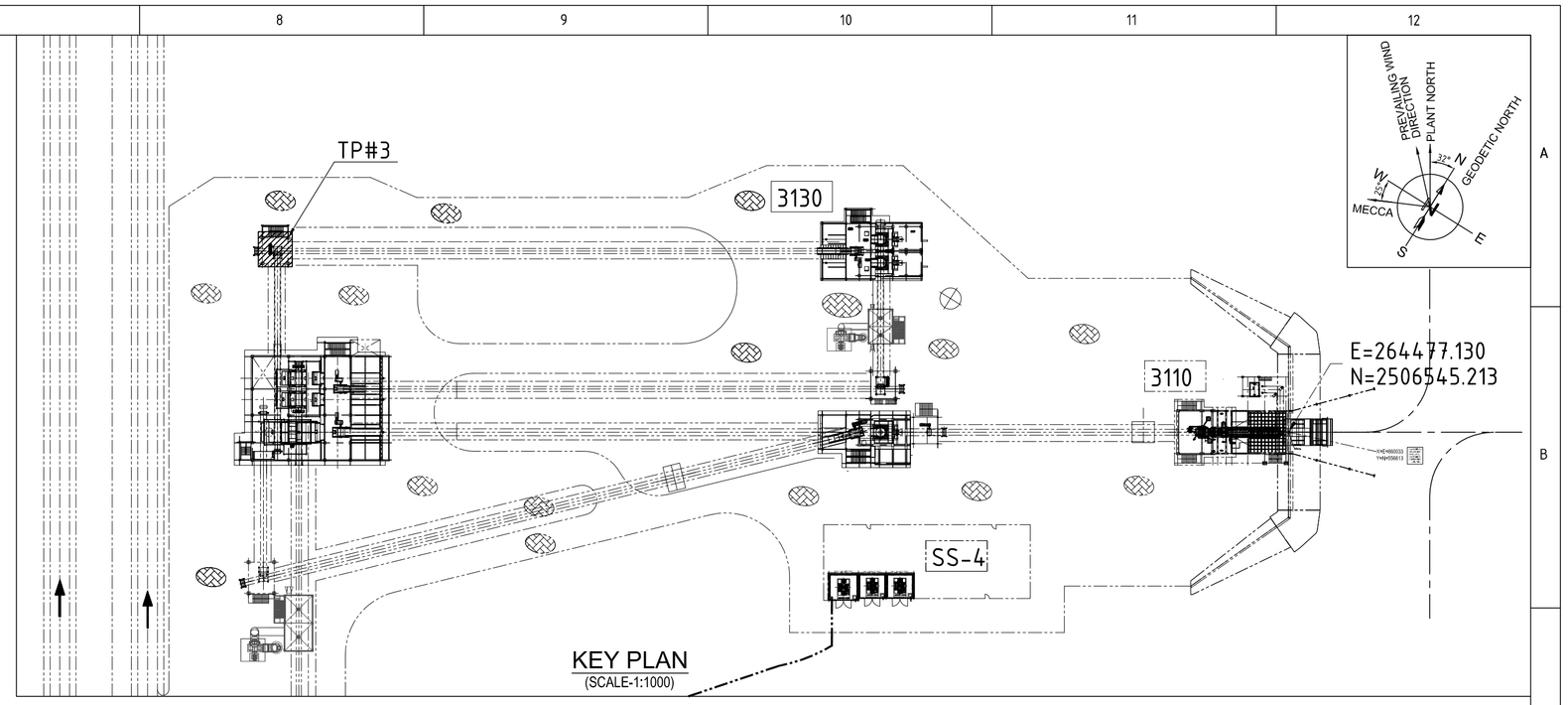
PLAN 6-6
(SCALE-1:50)



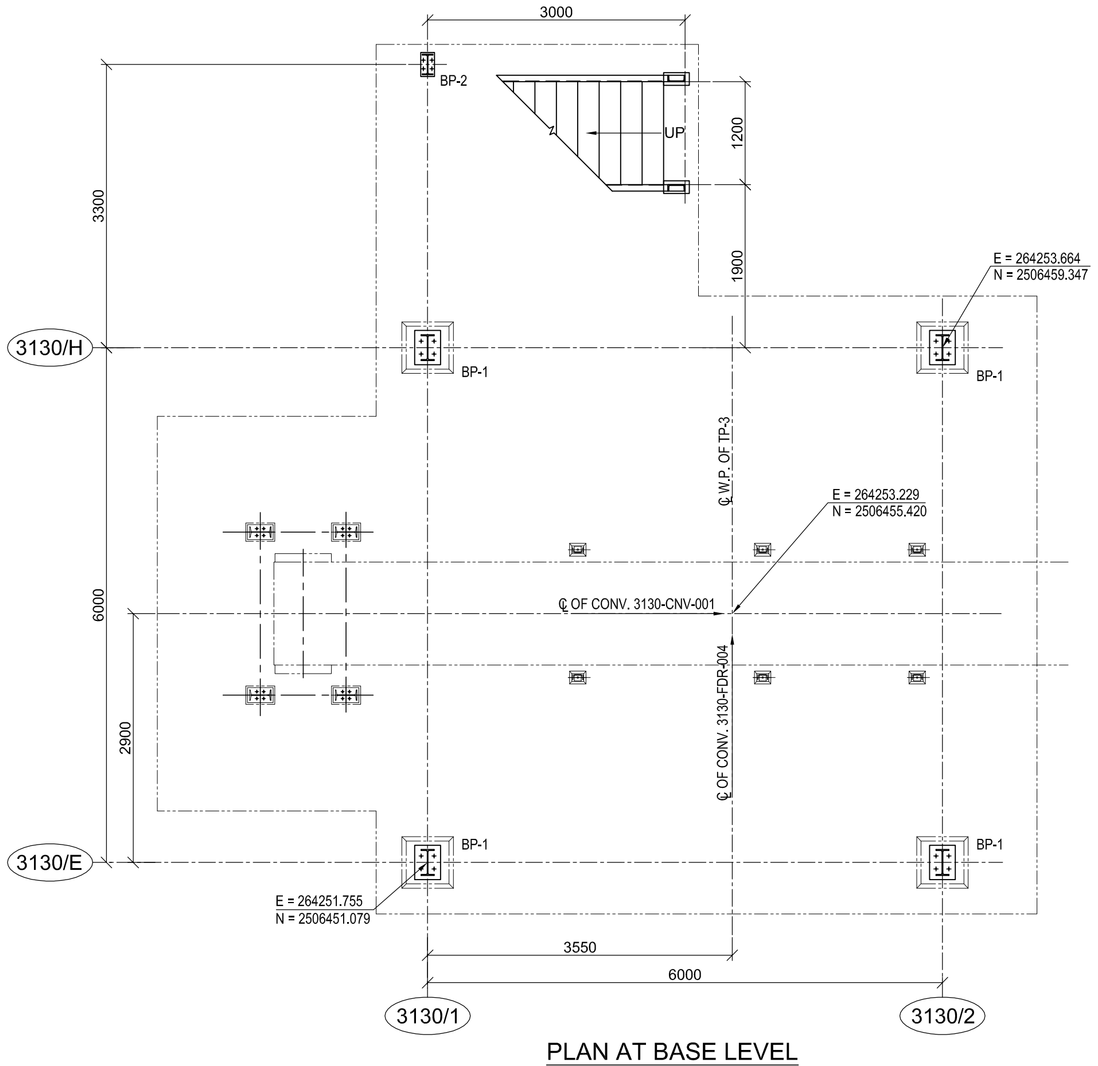
DETAIL 'E'
(SCALE - 1:10)



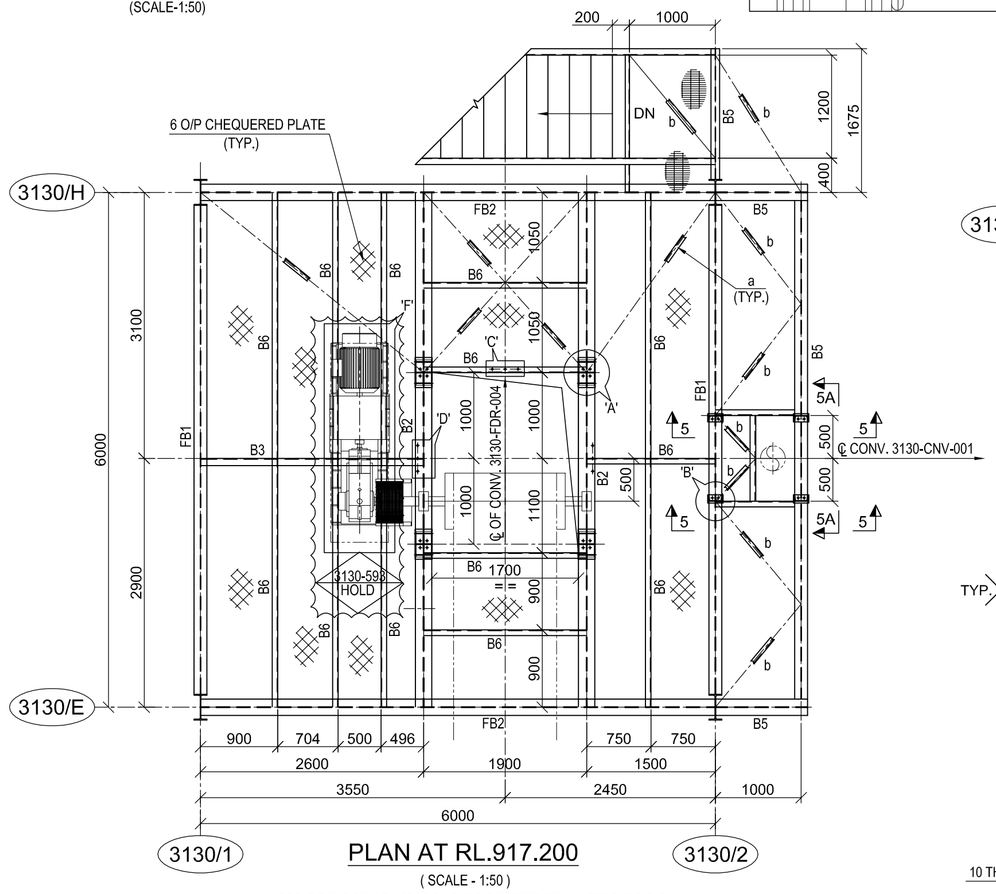
SECTION E1-E1
(SCALE - 1:10)



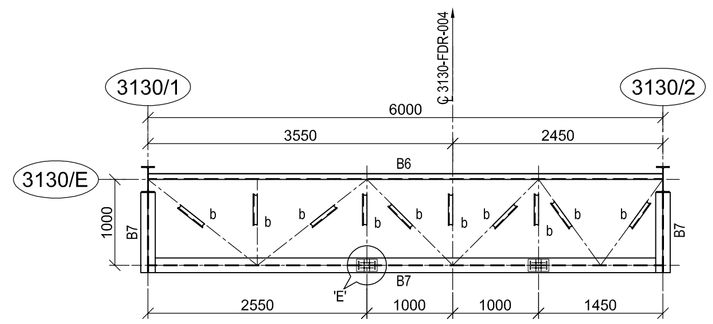
KEY PLAN
(SCALE-1:1000)



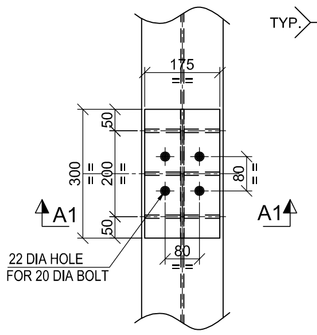
PLAN AT BASE LEVEL
(U/S OF BASE PLATE AT RL. 913.500 UNLESS NOTED OTHER WISE)
(SCALE - 1:50)



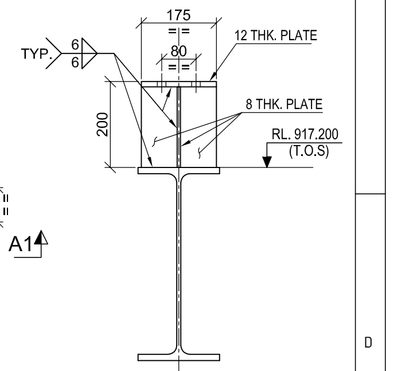
PLAN AT RL.917.200
(SCALE - 1:50)
(TOP OF FLOOR BRACING SHOULD BE KEPT 175mm BELOW T.O.S.)



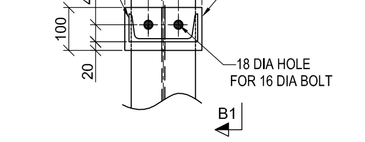
PLAN AT RL.915.321
(SCALE - 1:50)
(TOP OF FLOOR BRACING SHOULD BE KEPT 175mm BELOW T.O.S.)



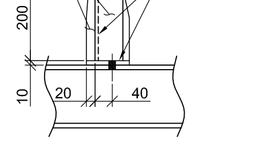
DETAIL 'A'
(SCALE - 1:10)



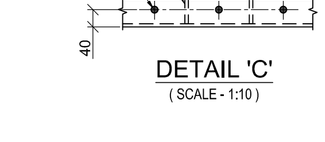
SECTION A1-A1
(SCALE - 1:10)



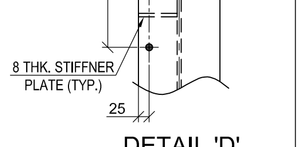
DETAIL 'B'
(SCALE - 1:10)



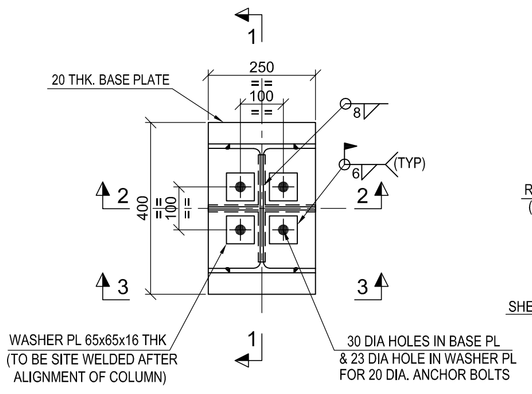
SECTION B1-B1
(SCALE - 1:10)



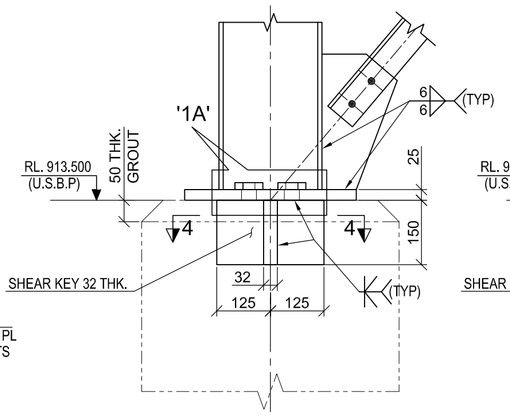
DETAIL 'C'
(SCALE - 1:10)



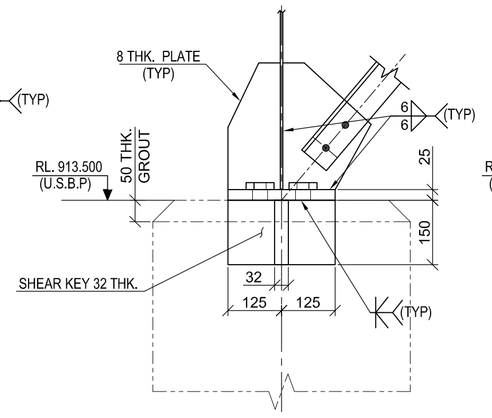
DETAIL 'D'
(SCALE - 1:10)



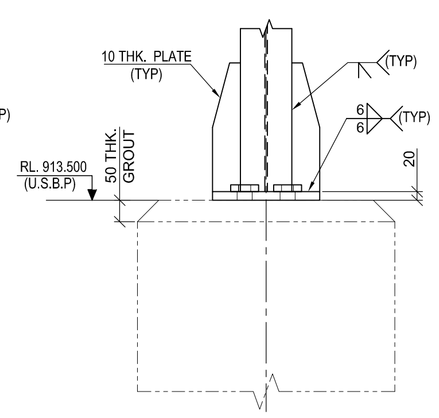
DETAIL OF BASE PLATE MKD. 'BP1'
(SCALE - 1:10)



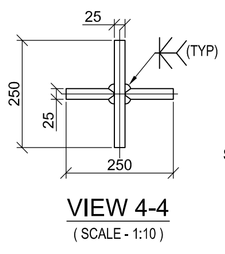
SECTION 1-1
(SCALE - 1:10)



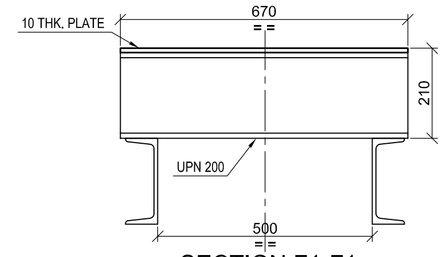
SECTION 2-2
(SCALE - 1:10)



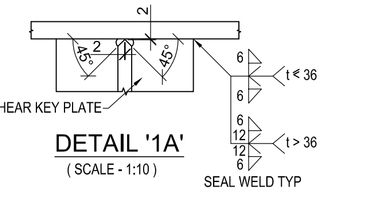
SECTION 3-3
(SCALE - 1:10)



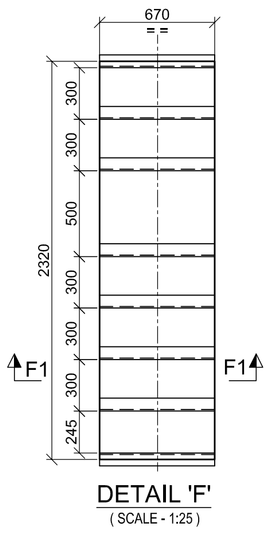
VIEW 4-4
(SCALE - 1:10)



SECTION F1-F1
(SCALE - 1:10)



DETAIL '1A'
(SCALE - 1:10)



DETAIL 'F'
(SCALE - 1:25)

LEGEND :-
R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING
   MD - 216 - 3130 -EG -ST - GAD - 1025
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO
   MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1015	TRANSFER POINT TP#3 - PLAN VIEW								
MD-216-3130-EG-ME-ELEV-1020	TRANSFER POINT TP#3 - ELEVATION	A01	06/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26205_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31311 — DOCUMENT ID: OU100668072 — REVISION: 0

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DRAWN	SUS	03/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	04/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	03/04/2020	STRUCTURAL GA OF TP-3(3130 CNV-001)-1
APPROVED	PRI	06/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE 1:100,20,15		A1	DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1024 — REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF TP-3 FOUNDATION**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 09-Apr-2020
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 1 0; 3 0 1 6; 5 6 1 0; 7 6 1 6; 14 0 -2 0; 15 0 -2 6; 16 6 -2 0; 17 6 -2 6;
18 0 0.5 0; 19 0 0.5 6; 20 6 0.5 6; 21 6 0.5 0;
MEMBER INCIDENCES
113 14 18; 114 15 19; 115 17 20; 116 16 21; 121 18 1; 122 19 3; 123 18 19;
124 20 7; 125 19 20; 126 21 5; 127 18 21; 128 21 20;
START GROUP DEFINITION
FLOOR
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
ISOTROPIC ALUMINUM
E 6.89476e+007
POISSON 0.33
DENSITY 26.6018
ALPHA 2.3e-005
DAMP 0.03
END DEFINE MATERIAL
MEMBER PROPERTY FRENCH
113 TO 116 121 122 124 126 PRIS YD 0.6 ZD 0.6
123 125 127 128 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
14 TO 17 FIXED
LOAD 1 DEAD  LOAD
JOINT LOAD
1 FX -1.7 FY -13.7 FZ 2
3 FX -0.1 FY -57 FZ -1.4
5 FX 1.7 FY -23.5 FZ 3
7 FX 3.9 FY -71.8 FZ -3.7
SELFWEIGHT Y -1
LOAD 2 LIVE LOAD
```

```
JOINT LOAD
1 FX -4.1 FY -38.4 FZ -5.3
3 FX -2.4 FY -57.1 FZ 5.2
5 FX 4 FY -52 FZ 0.3
7 FX 2.5 FY -59.6 FZ -0.2
LOAD 3 SDL-1 EQUIPMENT DEAD LOAD CASE -1
JOINT LOAD
1 FX -6.3 FY -22.4 FZ 22.6
3 FX -1.4 FY -129.1 FZ 33.6
5 FX 17.6 FY -38.9 FZ 31.7
7 FX 12.1 FY -180.7 FZ 41.1
LOAD 4 SDL-2 EQUIPMENT DEAD LOAD CASE -2
JOINT LOAD
1 FX -5.4 FY 1.3 FZ 24.2
3 FX -0.3 FY -119.5 FZ 29.6
5 FX 13.9 FY 2.3 FZ 36.1
7 FX 10.8 FY -166.1 FZ 36
LOAD 5 WIND LOAD (+X)
JOINT LOAD
1 FX 10.08 FY 13.86 FZ 1.63
3 FX 14.83 FY 16.44 FZ 3.45
5 FX 8.88 FY -13.86 FZ 2.05
7 FX 14.56 FY -16.44 FZ -7.14
LOAD 6 WIND LOAD (-X)
JOINT LOAD
1 FX -10.08 FY -13.86 FZ -1.63
3 FX -14.83 FY -16.44 FZ -3.45
5 FX -8.88 FY 13.86 FZ -2.05
7 FX -14.56 FY 16.44 FZ 7.14
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
1 FX 0.01 FY 6.35 FZ 6.26
3 FX -0.36 FY -6.35 FZ 6.67
5 FX 0.64 FY 13.35 FZ 10.44
7 FX -0.29 FY -13.35 FZ 11
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
1 FX -0.01 FY -6.35 FZ -6.26
3 FX 0.36 FY 6.35 FZ -6.67
5 FX -0.64 FY -13.35 FZ -10.44
7 FX 0.29 FY 13.35 FZ -11
LOAD 9 TEMPERATURE LOAD 1
JOINT LOAD
1 FX -61.27 FY -4.75 FZ -2.52
3 FX -39.14 FY 4.75 FZ -1.33
5 FX 57.42 FY 4.75 FZ 0.88
7 FX 42.99 FY -4.75 FZ 2.98
LOAD 10 TEMPERATURE LOAD 2
JOINT LOAD
1 FX 61.27 FY 4.75 FZ 2.52
3 FX 39.14 FY -4.75 FZ 1.33
5 FX -57.42 FY -4.75 FZ -0.88
7 FX -42.99 FY 4.75 FZ -2.98
LOAD 11 SEISMIC LOAD (+X)
```

```
JOINT LOAD
1 FX 2.33 FY 3.61 FZ 0.57
3 FX 2.51 FY 2.5 FZ 0.82
5 FX 2.08 FY -3.61 FZ 0.19
7 FX 2.44 FY -2.5 FZ -1.59
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
1 FX -0.25 FY 2.43 FZ 1.97
3 FX 0.02 FY -2.68 FZ 2.65
5 FX 0.29 FY 3.85 FZ 2.8
7 FX -0.07 FY -4.2 FZ 3.74
***LOAD COMBINATION FOR BASE PRESSURE CHECK
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
```

```
LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
```

```
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
```

```
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
```

```
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
LOAD LIST 401 TO 438
START CONCRETE DESIGN
CODE ACI 2008
FYMAIN 420000 ALL
FC 35000 ALL
MINMAIN 16 ALL
DESIGN COLUMN 113 TO 116 121 122 124 126
DESIGN BEAM 123 125 127 128
CLT 0.05 ALL
END CONCRETE DESIGN
FINISH
```

```
            **************************************************
            *                                                *
            *           STAAD.Pro CONNECT Edition            *
            *           Version  21.00.01.12                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    APR  9, 2020                *
            *           Time=    21:29: 6                    *
            *                                                *
            *  Licensed to: MMH KOLKATTA                     *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\TP 3\Civil\Staad\TP-3 Civil.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-2020
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 1 0; 3 0 1 6; 5 6 1 0; 7 6 1 6; 14 0 -2 0; 15 0 -2 6; 16 6 -2 0; 17 6 -2 6
     9. 18 0 0.5 0; 19 0 0.5 6; 20 6 0.5 6; 21 6 0.5 0
    10. MEMBER INCIDENCES
    11. 113 14 18; 114 15 19; 115 17 20; 116 16 21; 121 18 1; 122 19 3; 123 18 19
    12. 124 20 7; 125 19 20; 126 21 5; 127 18 21; 128 21 20
    13. START GROUP DEFINITION
    14. FLOOR
    15. END GROUP DEFINITION

   ** WARNING: NO VALID GROUPS IN GROUP DEFINITION DATA **
    16. DEFINE MATERIAL START
    17. ISOTROPIC STEEL
    18. E 2.05E+008
    19. POISSON 0.3
    20. DENSITY 76.8195
    21. ALPHA 1.2E-005
    22. DAMP 0.03
    23. TYPE STEEL
    24. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
    25. ISOTROPIC CONCRETE
    26. E 2.17185E+007
    27. POISSON 0.17
    28. DENSITY 23.5616
    29. ALPHA 1E-005
    30. DAMP 0.05
    31. TYPE CONCRETE
    32. STRENGTH FCU 27579
    33. ISOTROPIC ALUMINUM
    34. E 6.89476E+007
    35. POISSON 0.33
    36. DENSITY 26.6018
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37. ALPHA 2.3E-005
38. DAMP 0.03
39. END DEFINE MATERIAL
40. MEMBER PROPERTY FRENCH
41. 113 TO 116 121 122 124 126 PRIS YD 0.6 ZD 0.6
42. 123 125 127 128 PRIS YD 0.5 ZD 0.3
43. CONSTANTS
44. MATERIAL CONCRETE ALL
45. SUPPORTS
46. 14 TO 17 FIXED
47. LOAD 1 DEAD  LOAD
48. JOINT LOAD
49. 1 FX -1.7 FY -13.7 FZ 2
50. 3 FX -0.1 FY -57 FZ -1.4
51. 5 FX 1.7 FY -23.5 FZ 3
52. 7 FX 3.9 FY -71.8 FZ -3.7
53. SELFWEIGHT Y -1
54. LOAD 2 LIVE LOAD
55. JOINT LOAD
56. 1 FX -4.1 FY -38.4 FZ -5.3
57. 3 FX -2.4 FY -57.1 FZ 5.2
58. 5 FX 4 FY -52 FZ 0.3
59. 7 FX 2.5 FY -59.6 FZ -0.2
60. LOAD 3 SDL-1 EQUIPMENT DEAD LOAD CASE -1.
61. JOINT LOAD
62. 1 FX -6.3 FY -22.4 FZ 22.6
63. 3 FX -1.4 FY -129.1 FZ 33.6
64. 5 FX 17.6 FY -38.9 FZ 31.7
65. 7 FX 12.1 FY -180.7 FZ 41.1
66. LOAD 4 SDL-2 EQUIPMENT DEAD LOAD CASE -2
67. JOINT LOAD
68. 1 FX -5.4 FY 1.3 FZ 24.2
69. 3 FX -0.3 FY -119.5 FZ 29.6
70. 5 FX 13.9 FY 2.3 FZ 36.1
71. 7 FX 10.8 FY -166.1 FZ 36
72. LOAD 5 WIND LOAD (+X)
73. JOINT LOAD
74. 1 FX 10.08 FY 13.86 FZ 1.63
75. 3 FX 14.83 FY 16.44 FZ 3.45
76. 5 FX 8.88 FY -13.86 FZ 2.05
77. 7 FX 14.56 FY -16.44 FZ -7.14
78. LOAD 6 WIND LOAD (-X)
79. JOINT LOAD
80. 1 FX -10.08 FY -13.86 FZ -1.63
81. 3 FX -14.83 FY -16.44 FZ -3.45
82. 5 FX -8.88 FY 13.86 FZ -2.05
83. 7 FX -14.56 FY 16.44 FZ 7.14
84. LOAD 7 WIND LOAD (+Z)
85. JOINT LOAD
86. 1 FX 0.01 FY 6.35 FZ 6.26
87. 3 FX -0.36 FY -6.35 FZ 6.67
88. 5 FX 0.64 FY 13.35 FZ 10.44
89. 7 FX -0.29 FY -13.35 FZ 11
90. LOAD 8 WIND LOAD (-Z)
91. JOINT LOAD
92. 1 FX -0.01 FY -6.35 FZ -6.26
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 93. 3 FX 0.36 FY 6.35 FZ -6.67
 94. 5 FX -0.64 FY -13.35 FZ -10.44
 95. 7 FX 0.29 FY 13.35 FZ -11
 96. LOAD 9 TEMPERATURE LOAD 1
 97. JOINT LOAD
 98. 1 FX -61.27 FY -4.75 FZ -2.52
 99. 3 FX -39.14 FY 4.75 FZ -1.33
100. 5 FX 57.42 FY 4.75 FZ 0.88
101. 7 FX 42.99 FY -4.75 FZ 2.98
102. LOAD 10 TEMPERATURE LOAD 2
103. JOINT LOAD
104. 1 FX 61.27 FY 4.75 FZ 2.52
105. 3 FX 39.14 FY -4.75 FZ 1.33
106. 5 FX -57.42 FY -4.75 FZ -0.88
107. 7 FX -42.99 FY 4.75 FZ -2.98
108. LOAD 11 SEISMIC LOAD (+X)
109. JOINT LOAD
110. 1 FX 2.33 FY 3.61 FZ 0.57
111. 3 FX 2.51 FY 2.5 FZ 0.82
112. 5 FX 2.08 FY -3.61 FZ 0.19
113. 7 FX 2.44 FY -2.5 FZ -1.59
114. LOAD 12 SEISMIC LOAD (+Z)
115. JOINT LOAD
116. 1 FX -0.25 FY 2.43 FZ 1.97
117. 3 FX 0.02 FY -2.68 FZ 2.65
118. 5 FX 0.29 FY 3.85 FZ 2.8
119. 7 FX -0.07 FY -4.2 FZ 3.74
120. ***LOAD COMBINATION FOR BASE PRESSURE CHECK
121. LOAD 301 DL+SDL1
122. REPEAT LOAD
123. 1 1.0 3 1.0
124. LOAD 302 DL+SDL2
125. REPEAT LOAD
126. 1 1.0 4 1.0
127. LOAD 303 DL + SDL1 + TL1 + LL
128. REPEAT LOAD
129. 1 1.0 3 1.0 9 1.0 2 1.0
130. LOAD 304 DL + SDL1 + TL1 + LL
131. REPEAT LOAD
132. 1 1.0 4 1.0 9 1.0 2 1.0
133. LOAD 305 DL + SDL2 + TL2 + LL
134. REPEAT LOAD
135. 1 1.0 3 1.0 10 1.0 2 1.0
136. LOAD 306 DL + SDL2 + TL2 + LL
137. REPEAT LOAD
138. 1 1.0 4 1.0 10 1.0 2 1.0
139. LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
140. REPEAT LOAD
141. 1 1.0 3 1.0 2 0.75 5 0.75
142. LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
143. REPEAT LOAD
144. 1 1.0 3 1.0 2 0.75 6 0.75
145. LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
146. REPEAT LOAD
147. 1 1.0 3 1.0 2 0.75 7 0.75
148. LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
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149. REPEAT LOAD
150. 1 1.0 3 1.0 2 0.75 8 0.75
151. LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
152. REPEAT LOAD
153. 1 1.0 4 1.0 2 0.75 5 0.75
154. LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
155. REPEAT LOAD
156. 1 1.0 4 1.0 2 0.75 6 0.75
157. LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
158. REPEAT LOAD
159. 1 1.0 4 1.0 2 0.75 7 0.75
160. LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
161. REPEAT LOAD
162. 1 1.0 4 1.0 2 0.75 8 0.75
163. LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
164. REPEAT LOAD
165. 1 1.0 3 1.0 2 0.75 11 0.525
166. LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
167. REPEAT LOAD
168. 1 1.0 3 1.0 2 0.75 11 -0.525
169. LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
170. REPEAT LOAD
171. 1 1.0 3 1.0 2 0.75 12 0.525
172. LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
173. REPEAT LOAD
174. 1 1.0 3 1.0 2 0.75 12 -0.525
175. LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
176. REPEAT LOAD
177. 1 1.0 4 1.0 2 0.75 11 0.525
178. LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
179. REPEAT LOAD
180. 1 1.0 4 1.0 2 0.75 11 -0.525
181. LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
182. REPEAT LOAD
183. 1 1.0 4 1.0 2 0.75 12 0.525
184. LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
185. REPEAT LOAD
186. 1 1.0 4 1.0 2 0.75 12 -0.525
187. LOAD 323 0.6 (DL + SDL1) + WL(+X)
188. REPEAT LOAD
189. 1 0.6 3 0.6 5 1.0
190. LOAD 324 0.6 (DL + SDL1) + WL(-X)
191. REPEAT LOAD
192. 1 0.6 3 0.6 6 1.0
193. LOAD 325 0.6 (DL + SDL1) + WL(+Z)
194. REPEAT LOAD
195. 1 0.6 3 0.6 7 1.0
196. LOAD 326 0.6 (DL + SDL1) + WL(-Z)
197. REPEAT LOAD
198. 1 0.6 3 0.6 8 1.0
199. LOAD 327 0.6 (DL + SDL2) + WL(+X)
200. REPEAT LOAD
201. 1 0.6 4 0.6 5 1.0
202. LOAD 328 0.6 (DL + SDL2) + WL(-X)
203. REPEAT LOAD
204. 1 0.6 4 0.6 6 1.0
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205. LOAD 329 0.6 (DL + SDL2) + WL(+Z)
206. REPEAT LOAD
207. 1 0.6 4 0.6 7 1.0
208. LOAD 330 0.6 (DL + SDL2) + WL(-Z)
209. REPEAT LOAD
210. 1 0.6 4 0.6 8 1.0
211. LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
212. REPEAT LOAD
213. 1 0.6 3 0.6 11 0.7
214. LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
215. REPEAT LOAD
216. 1 0.6 3 0.6 11 -0.7
217. LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
218. REPEAT LOAD
219. 1 0.6 3 0.6 12 0.7
220. LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
221. REPEAT LOAD
222. 1 0.6 3 0.6 12 -0.7
223. LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
224. REPEAT LOAD
225. 1 0.6 4 0.6 11 0.7
226. LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
227. REPEAT LOAD
228. 1 0.6 4 0.6 11 -0.7
229. LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
230. REPEAT LOAD
231. 1 0.6 4 0.6 12 0.7
232. LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
233. REPEAT LOAD
234. 1 0.6 4 0.6 12 -0.7
235. ****
236. *LOAD COMBINATION FOR CONCRETE DESIGN
237. LOAD 401 1.4 (DL+SDL1)
238. REPEAT LOAD
239. 1 1.4 3 1.4
240. LOAD 402 1.4 (DL+SDL2)
241. REPEAT LOAD
242. 1 1.4 4 1.4
243. LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
244. REPEAT LOAD
245. 1 1.4 3 1.4 9 1.4 2 1.7
246. LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
247. REPEAT LOAD
248. 1 1.4 4 1.4 9 1.4 2 1.7
249. LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
250. REPEAT LOAD
251. 1 1.4 3 1.4 10 1.4 2 1.7
252. LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
253. REPEAT LOAD
254. 1 1.4 4 1.4 10 1.4 2 1.7
255. LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
256. REPEAT LOAD
257. 1 1.2 3 1.2 2 0.5 5 1.6
258. LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
259. REPEAT LOAD
260. 1 1.2 3 1.2 2 0.5 6 1.6
```

```
261. LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
262. REPEAT LOAD
263. 1 1.2 3 1.2 2 0.5 7 1.6
264. LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
265. REPEAT LOAD
266. 1 1.2 3 1.2 2 0.5 8 1.6
267. LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
268. REPEAT LOAD
269. 1 1.2 4 1.2 2 0.5 5 1.6
270. LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
271. REPEAT LOAD
272. 1 1.2 4 1.2 2 0.5 6 1.6
273. LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
274. REPEAT LOAD
275. 1 1.2 4 1.2 2 0.5 7 1.6
276. LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
277. REPEAT LOAD
278. 1 1.2 4 1.2 2 0.5 8 1.6
279. LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
280. REPEAT LOAD
281. 1 1.2 3 1.2 2 0.5 11 1.0
282. LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
283. REPEAT LOAD
284. 1 1.2 3 1.2 2 0.5 11 -1.0
285. LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
286. REPEAT LOAD
287. 1 1.2 3 1.2 2 0.5 12 1.0
288. LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
289. REPEAT LOAD
290. 1 1.2 3 1.2 2 0.5 12 -1.0
291. LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
292. REPEAT LOAD
293. 1 1.2 4 1.2 2 0.5 11 1.0
294. LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
295. REPEAT LOAD
296. 1 1.2 4 1.2 2 0.5 11 -1.0
297. LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
298. REPEAT LOAD
299. 1 1.2 4 1.2 2 0.5 12 1.0
300. LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
301. REPEAT LOAD
302. 1 1.2 4 1.2 2 0.5 12 -1.0
303. LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
304. REPEAT LOAD
305. 1 0.9 3 0.9 5 1.6
306. LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
307. REPEAT LOAD
308. 1 0.9 3 0.9 6 1.6
309. LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
310. REPEAT LOAD
311. 1 0.9 3 0.9 7 1.6
312. LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
313. REPEAT LOAD
314. 1 0.9 3 0.9 8 1.6
315. LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
316. REPEAT LOAD
```

```
317. 1 0.9 4 0.9 5 1.6
318. LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
319. REPEAT LOAD
320. 1 0.9 4 0.9 6 1.6
321. LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
322. REPEAT LOAD
323. 1 0.9 4 0.9 7 1.6
324. LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
325. REPEAT LOAD
326. 1 0.9 4 0.9 8 1.6
327. LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
328. REPEAT LOAD
329. 1 0.9 3 0.9 11 1.0
330. LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
331. REPEAT LOAD
332. 1 0.9 3 0.9 11 -1.0
333. LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
334. REPEAT LOAD
335. 1 0.9 3 0.9 12 1.0
336. LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
337. REPEAT LOAD
338. 1 0.9 3 0.9 12 -1.0
339. LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
340. REPEAT LOAD
341. 1 0.9 4 0.9 11 1.0
342. LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
343. REPEAT LOAD
344. 1 0.9 4 0.9 11 -1.0
345. LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
346. REPEAT LOAD
347. 1 0.9 4 0.9 12 1.0
348. LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
349. REPEAT LOAD
350. 1 0.9 4 0.9 12 -1.0
351. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 12 NUMBER OF MEMBERS 12
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 4

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 88, TOTAL DEGREES OF FREEDOM = 48
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
352. LOAD LIST 401 TO 438
353. START CONCRETE DESIGN
354. CODE ACI 2008
355. FYMAIN 420000 ALL
356. FC 35000 ALL
357. MINMAIN 16 ALL
358. DESIGN COLUMN 113 TO 116 121 122 124 126
```

==========

COLUMN NO. 113 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 114 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 115 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 116 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 121 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 122 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 124 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 126 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

**********END OF COLUMN DESIGN RESULTS**********

359. DESIGN BEAM 123 125 127 128

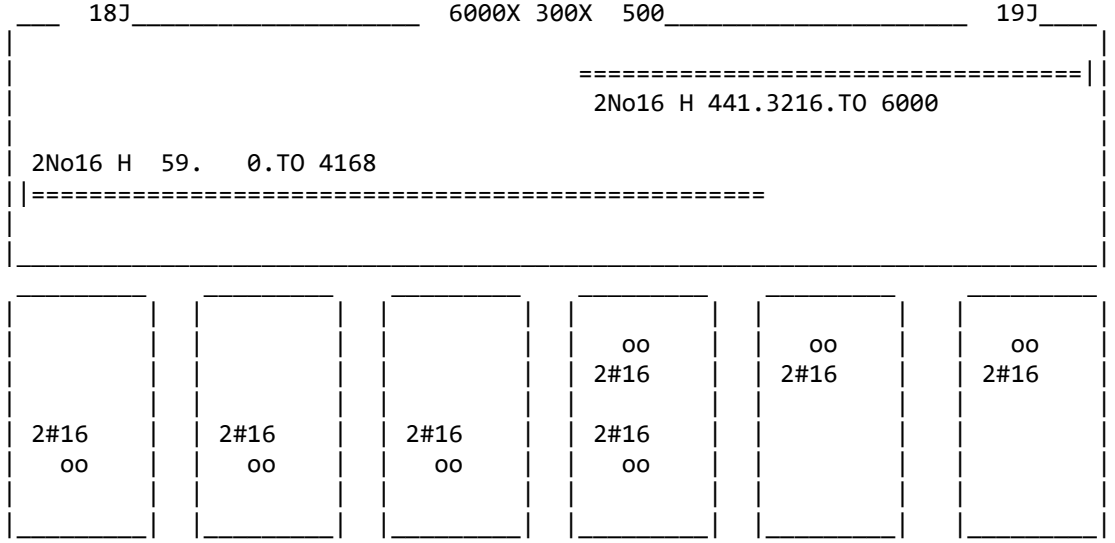
BEAM NO. 123 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	4168.	YES	NO
2	441.	2 - 16MM	3216.	6000.	NO	YES

B E A M N O. 123 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.11 KNS Vc= 123.98 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.26 KNS Vc= 124.60 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



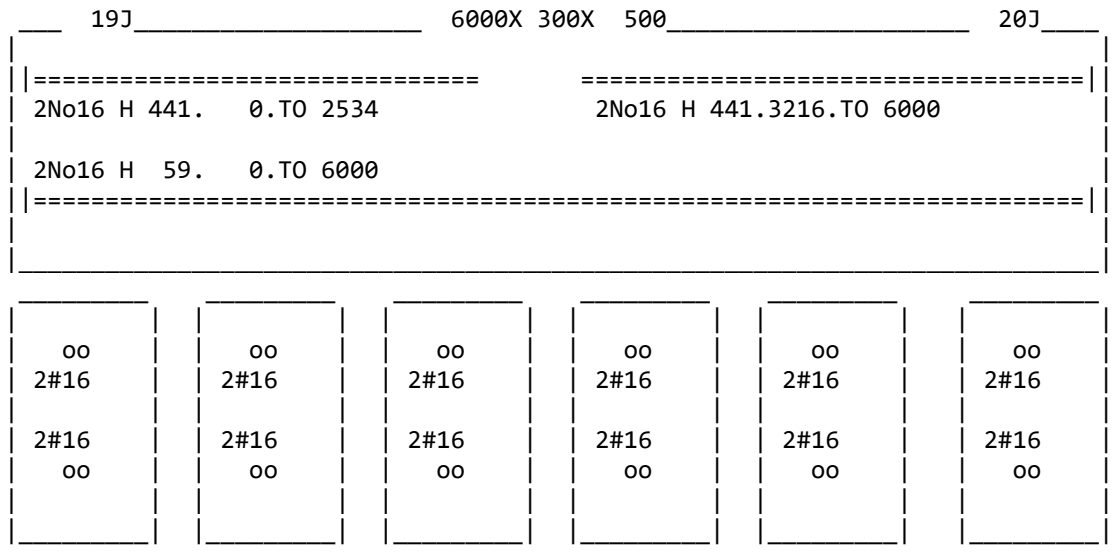
BEAM NO. 125 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	6000.	YES	YES
2	441.	2 - 16MM	0.	2534.	YES	NO
3	441.	2 - 16MM	3216.	6000.	NO	YES

B E A M N O. 125 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.33 KNS Vc= 128.42 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.04 KNS Vc= 128.42 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



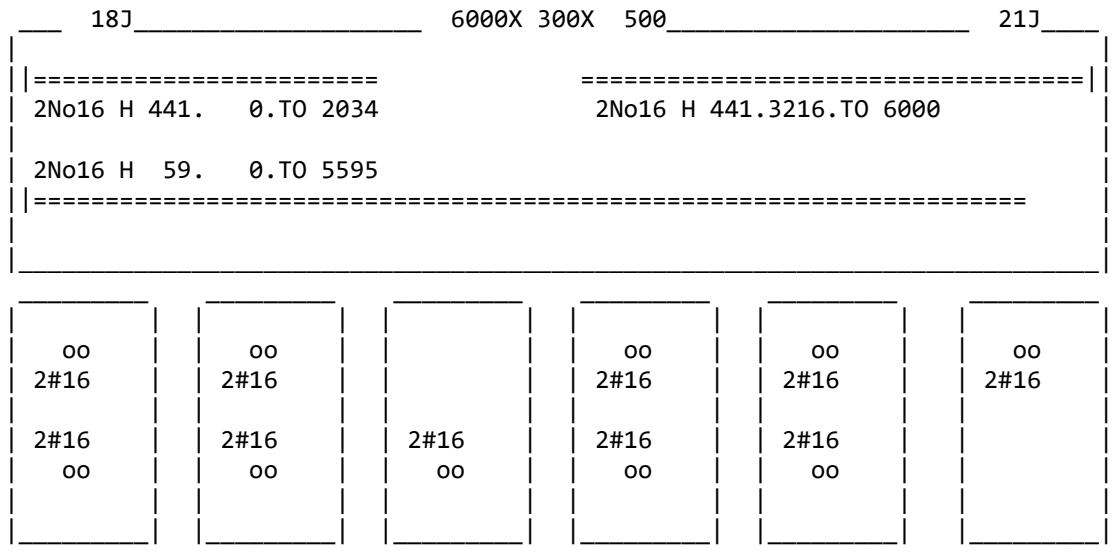
BEAM NO. 127 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5595.	YES	NO
2	441.	2 - 16MM	0.	2034.	YES	NO
3	441.	2 - 16MM	3216.	6000.	NO	YES

B E A M N O. 127 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.76 KNS Vc= 126.35 KNS Vs= 0.00 KNS
Tu= 0.50 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.60 KNS Vc= 126.35 KNS Vs= 0.00 KNS
Tu= 0.50 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



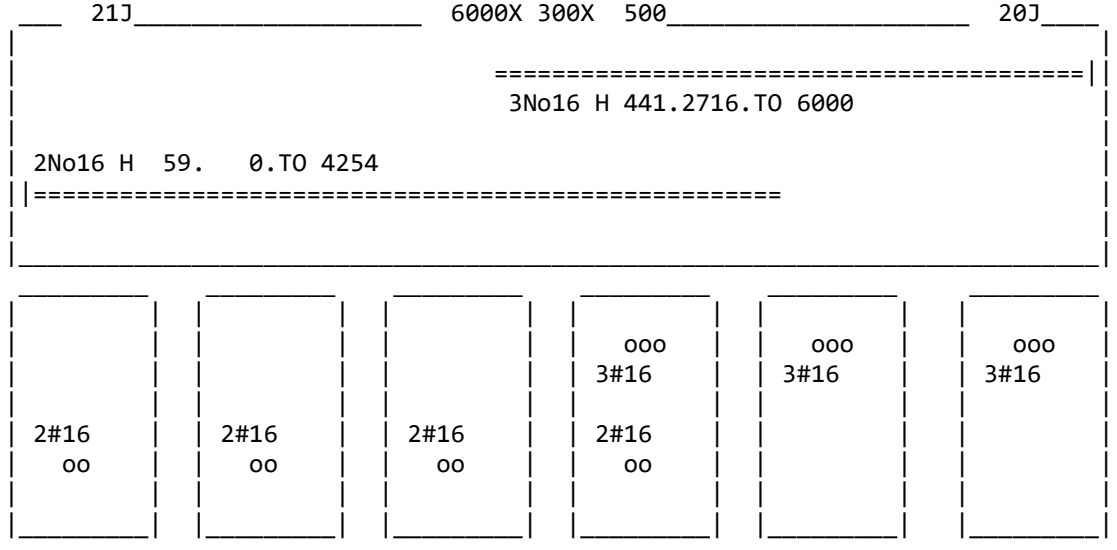
BEAM NO. 128 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	4254.	YES	NO
2	441.	3 - 16MM	2716.	6000.	NO	YES

B E A M N O. 128 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.77 KNS Vc= 123.80 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 24.60 KNS Vc= 124.98 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

360. CLT 0.05 ALL
361. END CONCRETE DESIGN

362. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 9,2020 TIME= 21:29: 8 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	10-04-2020	11-05-2020	01-11-2023				Date	10-04-2020	11-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **COMMENT RESOLUTION SHEET**

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002343, 5 May 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 5 May 2020	Comment by	EPC Response DATE: 11 May 2020	Status	Remarks
			General Comments (cc)				
MD-216-3130-EG-CV-CAL-1005-A01	3130	1	see comment on Design of TP-1	Mr. Doron Golan Bechtel	The description has been rectified as per comment in TP-1	Closed	